

Gap Analysis

Primary project source: ProofRoad v2, SIH 2026 PS 26007, uploaded by the user. The document describes ProofRoad as an engineering proposal and explicitly says it is not a safety certification. Page references below refer to the uploaded ProofRoad v2 document.

1. Problem-to-capability gap

Need	Common limitation	ProofRoad response	Evidence still needed
Operate when driver visibility is 3–5 m	Visual-only driving loses range	Radar-first proof	Mine-grade radar field data
Avoid false-clear decisions	Silence/timers can be misread	Unknown never clear; stale messages	Long-duration field validation
Use fleet information safely	Peer data may be stale/false	Age counters + signatures + local checks	Production key management
Avoid warning fatigue	Too many alerts are ignored	Speed-conditional + escalation-only alerts	Human-factors study
Maintain safety after failures	Cloud/network dependency	Local safety MCU	Fault-injection testing
Improve productivity	Stopping everything is inefficient	Slow only where proven distance is insufficient	Fleet simulation + field data

2. Research novelty claims to phrase carefully

Do not claim that radar, V2V, fog sensors or sensor fusion are individually novel. ProofRoad itself says these technologies are not new alone; the proposed contribution is their combination and the decision rules around freshness, proof, conservative fusion and replayability (p. 14).

Potentially distinctive engineering concepts include: proof freshness using age counters instead of a shared clock; hazard clearing by an actual observing pass rather than a timer; right-of-way based on remaining room to stop; fog-post audit by passing trucks; and a virtual-fleet simulation using the same message model (pp. 1 and 15).

Any patent claim would require a separate prior-art search. The current proposal does not establish patentability.

3. Main unresolved gaps

Mine-grade radar performance under local fog/rain/dust.

Representative braking data and validated safe-speed envelope.

Real haul-road radio behavior.

Real-world driver interaction and warning response.

Production-grade device identity, secure boot, signed firmware and key lifecycle.

Validation of the productivity model against actual haul-cycle data.